

CS253 - Data Structures and Algorithms for DS/AI

Spring 2025

PSO 6, Week 7

Question 1

Let $S = 69, 79, 122, 72, 50, 14, 140, 134$ be a sequence of numbers, and let $h_1(k) = k \bmod 13$ and $h_2(k) = 1 + (k \bmod 11)$. Insert S into a hash table of size 13 using double hashing with h_1 and h_2 .

Question 2

Fill in the following table with the asymptotic order of the runtime for each operation in the **worst** case. Your hash table answers should assume independent and uniform hashing (I.U.H.). GetSuccessor takes as input a reference to an element in the data structure and finds the next largest element.

	HT with Separate Chaining	HT with Open Addressing	BST	Balanced BST
Insert				
Search				
Delete				
GetMax				
GetSuccessor(x)				

Question 3

Prove that, assuming independent and uniform hashing, the expected number of probes in an unsuccessful search in a hash table is $\frac{1}{1-\alpha}$ where $\alpha = n/m$ is the load factor of the table.